

0 7

A meadow is an area of grassland with a wide range of plant and animal species. A student investigated whether cutting some of the plants in a meadow had any effect on the biodiversity of insects in that meadow.

The student created two sample areas, called plots, in the meadow. Each plot measured 10 m × 5 m

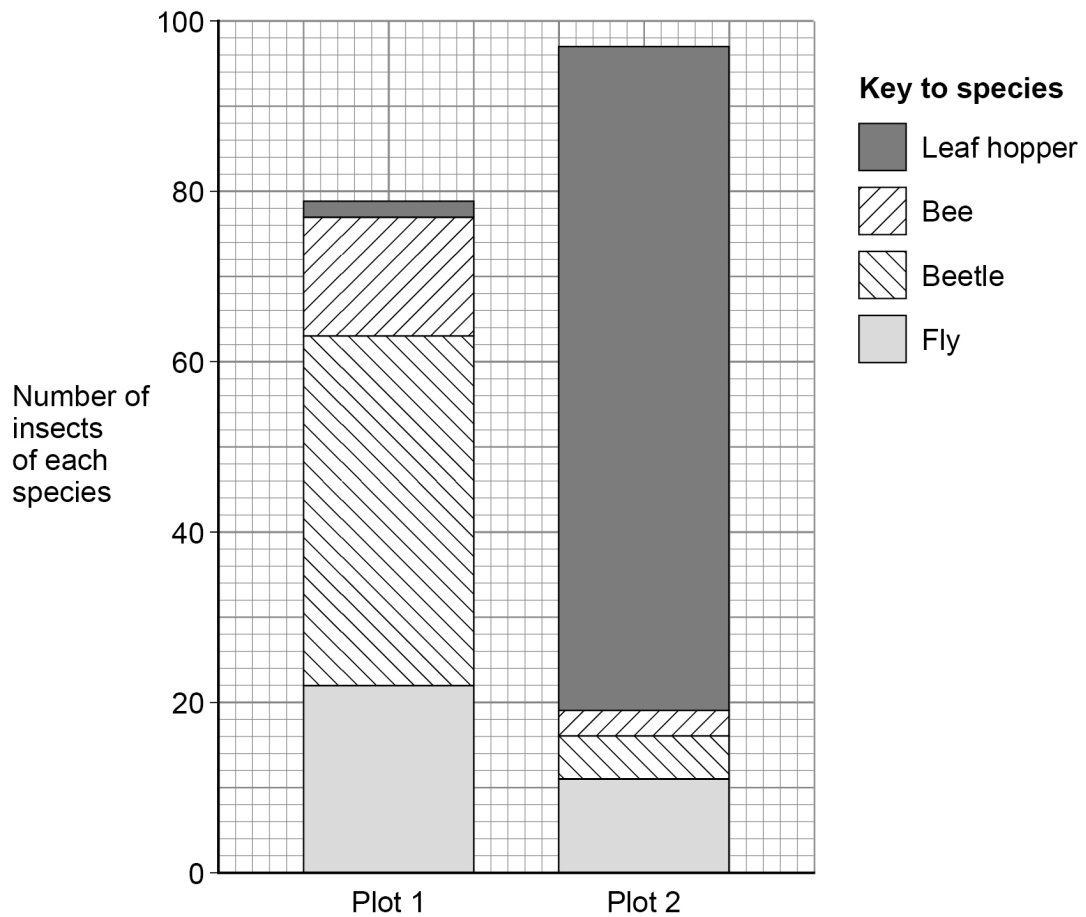
The student:

- did not cut plants in **plot 1**
- cut the plants in **plot 2** with a lawn mower once a week.

After 10 weeks, the student captured all of the organisms of four insect species found in each of these plots.

Figure 7 shows the student's results.

Figure 7



07.1

Use the information in **Figure 7** to calculate the index of diversity for the insects captured in **plot 1**.

The formula to calculate the index of diversity (d) is

$$d = \frac{N(N-1)}{\sum n(n-1)}$$

where N is the total number of insects of all species and n is the total number of insects of each species.

Give the answer to **2** significant figures and show your working.

[2 marks]

d _____

07.2

The student concluded that cutting plants with a lawn mower increased the species richness of insects in that meadow.

Use information in **Figure 7** to explain why the student's conclusion is incorrect.

[1 mark]

Question 7 continues on the next page

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[4 marks]

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7

